

set

May 15, 2026

```
[ ]: # LIST DATASTRUCTURE
```

```
[1]: l = []  
     l
```

```
[1]: []
```

```
[2]: type(l)
```

```
[2]: list
```

```
[3]: l.append(10)
```

```
[4]: l
```

```
[4]: [10]
```

```
[5]: l.append(20)  
     l.append(30)  
     l.append(40)  
     l.append(50)
```

```
[6]: l
```

```
[6]: [10, 20, 30, 40, 50]
```

```
[7]: len(l)
```

```
[7]: 5
```

```
[8]: l.append(10)
```

```
[9]: l
```

```
[9]: [10, 20, 30, 40, 50, 10]
```

```
[10]: l1 = [10, 2.3, 1+2j, True, 'nit', [1,2,3]]  
      l1
```

```
[10]: [10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
```

```
[11]: len(l1)
```

```
[11]: 6
```

```
[12]: print(l)
      print(l1)
```

```
[10, 20, 30, 40, 50, 10]
[10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
```

```
[14]: print(id(l))
      print(id(l1))
```

```
1864316344192
1864316254080
```

```
[15]: 1
```

```
[15]: [10, 20, 30, 40, 50, 10]
```

```
[16]: l2 = l.copy()
      l2
```

```
[16]: [10, 20, 30, 40, 50, 10]
```

```
[17]: print(id(l))
      print(id(l1))
      print(id(l2))
```

```
1864316344192
1864316254080
1864316273472
```

```
[18]: l2
```

```
[18]: [10, 20, 30, 40, 50, 10]
```

```
[19]: len(l2)
```

```
[19]: 6
```

```
[20]: l2.clear()
```

```
[21]: len(l2)
```

```
[21]: 0
```

```
[22]: 12
```

```
[22]: []
```

```
[23]: del 12
```

```
[24]: 12
```

```
-----  
NameError                                Traceback (most recent call last)  
Cell In[24], line 1  
----> 1 12  
  
NameError: name '12' is not defined
```

```
[25]: 1
```

```
[25]: [10, 20, 30, 40, 50, 10]
```

```
[26]: l.count(10)
```

```
[26]: 2
```

```
[27]: l.count(100)
```

```
[27]: 0
```

```
[28]: print(1)  
      print(11)
```

```
[10, 20, 30, 40, 50, 10]  
[10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
```

```
[29]: a = [1,2,3]  
      b = [4,5,6]  
  
      a.extend(b)
```

```
[30]: a
```

```
[30]: [1, 2, 3, 4, 5, 6]
```

```
[31]: 12 = []  
      12
```

```
[31]: []
```

```
[32]: 12.extend(1)
```

```
[33]: 12
```

```
[33]: [10, 20, 30, 40, 50, 10]
```

```
[34]: 12.extend(11)
```

```
[35]: 12
```

```
[35]: [10, 20, 30, 40, 50, 10, 10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
```

```
[36]: print(1)
      print(11)
      print(12)
```

```
[10, 20, 30, 40, 50, 10]
```

```
[10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
```

```
[10, 20, 30, 40, 50, 10, 10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
```

```
[37]: 1
```

```
[37]: [10, 20, 30, 40, 50, 10]
```

```
[38]: 1[0]
```

```
[38]: 10
```

```
[39]: 1[0] = 100
```

```
[40]: 1
```

```
[40]: [100, 20, 30, 40, 50, 10]
```

```
[41]: job = [5000]
      job
```

```
[41]: [5000]
```

```
[42]: job[0]
```

```
[42]: 5000
```

```
[43]: job[0] = 10000
```

```
[44]: job
```

[44]: [10000]

[45]: 1

[45]: [100, 20, 30, 40, 50, 10]

[46]: 1[-2] = 500
1

[46]: [100, 20, 30, 40, 500, 10]

[47]: 1

[47]: [100, 20, 30, 40, 500, 10]

[48]: 1.index(100)

[48]: 0

[49]: 12

[49]: [10, 20, 30, 40, 50, 10, 10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]

[50]: 1.insert(2, 25)
1

[50]: [100, 20, 25, 30, 40, 500, 10]

[51]: 1

[51]: [100, 20, 25, 30, 40, 500, 10]

[52]: 12

[52]: [10, 20, 30, 40, 50, 10, 10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]

[53]: 12.pop()

[53]: [1, 2, 3]

[54]: 12

[54]: [10, 20, 30, 40, 50, 10, 10, 2.3, (1+2j), True, 'nit']

[55]: 12.pop()

[55]: 'nit'

```
[56]: 12.pop()
[56]: True
[57]: 12
[57]: [10, 20, 30, 40, 50, 10, 10, 2.3, (1+2j)]
[58]: 12.pop(2)
[58]: 30
[59]: 12
[59]: [10, 20, 40, 50, 10, 10, 2.3, (1+2j)]
[60]: 12.remove(40)
[61]: 12
[61]: [10, 20, 50, 10, 10, 2.3, (1+2j)]
[62]: 12.pop(4)
[62]: 10
[63]: 1
[63]: [100, 20, 25, 30, 40, 500, 10]
[64]: 1
[64]: [100, 20, 25, 30, 40, 500, 10]
[65]: 1.reverse()
[66]: 1
[66]: [10, 500, 40, 30, 25, 20, 100]
[67]: 1.sort() #ascending order
[68]: 1
[68]: [10, 20, 25, 30, 40, 100, 500]
```

```
[69]: l.sort(reverse=True)
1
```

```
[69]: [500, 100, 40, 30, 25, 20, 10]
```

```
[70]: l3 = [10,20,[2,3],3]
```

```
[71]: l3[2] = [2]
13
```

```
[71]: [10, 20, [2], 3]
```

```
[72]: 1
```

```
[72]: [500, 100, 40, 30, 25, 20, 10]
```

```
[73]: 5000 in l
```

```
[73]: False
```

```
[74]: 1
```

```
[74]: [500, 100, 40, 30, 25, 20, 10]
```

```
[75]: for i in l:
      print(i)
```

```
500
100
40
30
25
20
10
```

```
[76]: l3 = [3, 100, 2, 50]
13
```

```
[76]: [3, 100, 2, 50]
```

```
[77]: l3.sort()
```

```
[78]: 13
```

```
[78]: [2, 3, 50, 100]
```

```
[79]: 13.sort(reverse = True)
13
```

```
[79]: [100, 50, 3, 2]
```

```
[80]: 14 = [2, 'a', 2.4, True]
14
```

```
[80]: [2, 'a', 2.4, True]
```

```
[81]: 14.sort()
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[81], line 1
----> 1 14.sort()

TypeError: '<' not supported between instances of 'str' and 'int'
```

```
[82]: 15 = ['z', 'b', 'm', 'c']
15.sort() # parameter
15
```

```
[82]: ['b', 'c', 'm', 'z']
```

```
[83]: 15.sort(reverse= True)
```

```
[84]: 15
```

```
[84]: ['z', 'm', 'c', 'b']
```

```
[85]: # list indexing & slicing
```

```
[86]: 11
```

```
[86]: [10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
```

```
[87]: 11[:]
```

```
[87]: [10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
```

```
[88]: 11[::-2]
```

```
[88]: [[1, 2, 3], True, 2.3]
```

```
[89]: 11
```



```
[89]: [10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
```

```
[90]: l1[0:10:3]
```

```
[90]: [10, True]
```

```
[91]: l1
```

```
[91]: [10, 2.3, (1+2j), True, 'nit', [1, 2, 3]]
```

```
[92]: l1[2] = 45  
l1
```

```
[92]: [10, 2.3, 45, True, 'nit', [1, 2, 3]]
```

```
[93]: # list datastrucutre we are completed
```

```
[94]: # tuple or tuple
```

```
[95]: t = ()  
t
```

```
[95]: ()
```

```
[96]: type(t)
```

```
[96]: tuple
```

```
[97]: t1 = (10, 20, 30)  
t1
```

```
[97]: (10, 20, 30)
```

```
[98]: t1.count(10)
```

```
[98]: 1
```

```
[99]: t1.count(100)
```

```
[99]: 0
```

```
[100]: t1
```

```
[100]: (10, 20, 30)
```

```
[101]: t1.index(10)
```

```
[101]: 0
```

```
[102]: t1
```

```
[102]: (10, 20, 30)
```

```
[103]: t1[0] = 100
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[103], line 1  
----> 1 t1[0] = 100  
  
TypeError: 'tuple' object does not support item assignment
```

```
[104]: sbi = (1234, 'cizps')  
sbi
```

```
[104]: (1234, 'cizps')
```

```
[105]: sbi[0] = 456  
sbi
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[105], line 1  
----> 1 sbi[0] = 456  
      2 sbi  
  
TypeError: 'tuple' object does not support item assignment
```

```
[106]: sbi
```

```
[106]: (1234, 'cizps')
```

```
[107]: l
```

```
[107]: [500, 100, 40, 30, 25, 20, 10]
```

```
[108]: t1
```

```
[108]: (10, 20, 30)
```

```
[109]: t1 * 2
```

```
[109]: (10, 20, 30, 10, 20, 30)
```

```
[110]: # list & tuple is completed
```

```
[111]: l7= [10, 20, 40]
```

```
[112]: l7 * 2
```

```
[112]: [10, 20, 40, 10, 20, 40]
```

```
[113]: t1
```

```
[113]: (10, 20, 30)
```

```
[ ]: # SET
```

```
[114]: s = {}  
s
```

```
[114]: {}
```

```
[115]: type(s)
```

```
[115]: dict
```

```
[116]: s1 = set()  
s1
```

```
[116]: set()
```

```
[117]: s1 = {20, 30, 90, 10}  
s1
```

```
[117]: {10, 20, 30, 90}
```

```
[118]: type(s1)
```

```
[118]: set
```

```
[119]: s1
```

```
[119]: {10, 20, 30, 90}
```

```
[120]: s1.add(100)
```

```
[121]: s1
```

```
[121]: {10, 20, 30, 90, 100}
```

```
[124]: s1.add(15)
```

```
[123]: s1
```

```
[123]: {10, 15, 20, 30, 90, 100}
```

```
[125]: s1.add(5)
s1
```

```
[125]: {5, 10, 15, 20, 30, 90, 100}
```

```
[126]: s2 = {5, 5.5, 'nit', 1+2j, True}
s2
```

```
[126]: {(1+2j), 5, 5.5, True, 'nit'}
```

```
[127]: s2.add(100)
```

```
[128]: s2
```

```
[128]: {(1+2j), 100, 5, 5.5, True, 'nit'}
```

```
[129]: print(s1)
print(s2)
```

```
{100, 5, 10, 15, 20, 90, 30}
{True, 'nit', 5.5, 5, (1+2j), 100}
```

```
[130]: s3 = s2.copy()
```

```
[131]: s3
```

```
[131]: {(1+2j), 100, 5, 5.5, True, 'nit'}
```

```
[132]: s3[0]
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[132], line 1
----> 1 s3[0]

TypeError: 'set' object is not subscriptable
```

```
[133]: s3
```

```
[133]: {(1+2j), 100, 5, 5.5, True, 'nit'}
```

```
[134]: s3[:]
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[134], line 1  
----> 1 s3[:]  
  
TypeError: 'set' object is not subscriptable
```

```
[135]: s3
```

```
[135]: {(1+2j), 100, 5, 5.5, True, 'nit'}
```

```
[136]: s3.pop()
```

```
[136]: True
```

```
[137]: s3
```

```
[137]: {(1+2j), 100, 5, 5.5, 'nit'}
```

```
[138]: s3.pop(3)
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[138], line 1  
----> 1 s3.pop(3)  
  
TypeError: set.pop() takes no arguments (1 given)
```

```
[139]: s2
```

```
[139]: {(1+2j), 100, 5, 5.5, True, 'nit'}
```

```
[140]: s1
```

```
[140]: {5, 10, 15, 20, 30, 90, 100}
```

```
[141]: s2
```

```
[141]: {(1+2j), 100, 5, 5.5, True, 'nit'}
```

```
[142]: s2
```

```
[142]: {(1+2j), 100, 5, 5.5, True, 'nit'}
```

```

[143]: s2.discard(1+2j)

[144]: s2

[144]: {100, 5, 5.5, True, 'nit'}

[145]: s2.discard(1000)

[146]: s2

[146]: {100, 5, 5.5, True, 'nit'}

[147]: s2.add(100)

[148]: s2

[148]: {100, 5, 5.5, True, 'nit'}

[149]: s2

[149]: {100, 5, 5.5, True, 'nit'}

[150]: # SET OPERATION -->

[151]: s7 = {1,2,3,4,5}
      s8 = {4,5,6,7,8}
      s9 = {8,9,10}

[152]: s7.union(s8)

[152]: {1, 2, 3, 4, 5, 6, 7, 8}

[153]: s7.union(s8, s9)

[153]: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

[154]: s8 | s9

[154]: {4, 5, 6, 7, 8, 9, 10}

[155]: s8 | s9 | s7

[155]: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

[156]: print(s7)
      print(s8)
      print(s9)

```

```
{1, 2, 3, 4, 5}
{4, 5, 6, 7, 8}
{8, 9, 10}
```

```
[157]: s7 = {1,2,3,4,5}
      s8 = {4,5,6,7,8}
      s9 = {8,9,10}
```

```
[158]: s7.intersection(s8)
```

```
[158]: {4, 5}
```

```
[159]: s7.intersection(s9)
```

```
[159]: set()
```

```
[160]: s8 & s9
```

```
[160]: {8}
```

```
[161]: s7 = {1,2,3,4,5}
      s8 = {4,5,6,7,8}
      s9 = {8,9,10}
```

```
[162]: s7.difference(s8)
```

```
[162]: {1, 2, 3}
```

```
[163]: s8 - s9
```

```
[163]: {4, 5, 6, 7}
```

```
[164]: s9 - s8
```

```
[164]: {9, 10}
```

```
[165]: s7 = {1,2,3,4,5}
      s8 = {4,5,6,7,8}
      s9 = {8,9,10}
```

```
[166]: s7.symmetric_difference(s8)
```

```
[166]: {1, 2, 3, 6, 7, 8}
```

```
[167]: s8 ^ s9
```

```
[167]: {4, 5, 6, 7, 9, 10}
```

```
[168]: s7 = {1,2,3,4,5}
      s8 = {4,5,6,7,8}
      s9 = {8,9,10}
```

```
[169]: s7.symmetric_difference_update(s8)
```

```
[173]: s7
```

```
[173]: {1, 2, 3, 6, 7, 8}
```

```
[171]: # superset , subset , disjoint
```

```
[172]: a = {1,2,3,4,5,6,7,8,9}
      b = {3,4,5,6,7,8}
      c = {10,20,30,40}
```

```
[174]: a.issuperset(b)
```

```
[174]: True
```

```
[175]: b.issubset(a)
```

```
[175]: True
```

```
[176]: print(a)
      print(b)
      print(c)
```

```
{1, 2, 3, 4, 5, 6, 7, 8, 9}
{3, 4, 5, 6, 7, 8}
{40, 10, 20, 30}
```

```
[177]: c.isdisjoint(a)
```

```
[177]: True
```

```
[178]: c.isdisjoint(b)
```

```
[178]: True
```

```
[179]: a1 = {1,2,3,4,5,6,7,8,9}
      b1 = {30,40,50,60,70,80}
      c1 = {10,20,30,40}
```

```
[180]: a1.issuperset(b1)
```

```
[180]: False
```



```
[181]: a1.isdisjoint(b1)
```

```
[181]: True
```